NEAR EARTH ORBIT NETWORK

NEON

SMBA Implementation Phase Draft RFP Industry Day

April 25, 2025

Ground Rules

- The NEON team cannot have any technical discussions with Industry Partners outside of full group discussions and one on ones
- Information provided to one Industry Partner must be provided to all Industry Partners
  - Questions will be posted anonymously
- The final chart deck from today will be posted on SAM.gov

Implementation Phase Draft RFP Overview Welcome

- We are using today to
  - Elaborate on the thinking behind the documents that were released with the SMBA Implementation Phase Draft Request for Proposal (RFP)
  - Point towards areas of interest for feedback
  - Answer questions Industry Partners might have
  - Foster discussion on the SMBA instrument, selections, and contract

The SMBA Implementation Phase Statement of Work (SOW), Contract Data Requirements List (CDRL), Performance Requirements Document (PRD), Instrument Mission Assurance Requirements (IMAR), Evaluation Criteria, sample contract, and other information included in the Draft RFP are available on SAM.gov at the following link: https://sam.gov/opp/34f2e39ac9c0458aa90cb6e02ac2ef96/view

Anything in the final SMBA Implementation Phase Request for Proposal (RFP) will supersede the information provided as part of this overview.

Introductions

- Near Earth Orbit Network (NEON) Team
- Industry Partners
  - BAE Systems
  - Northrop Grumman
  - L3Harris
  - Raytheon
  - Spire Global
  - Teledyne Brown Engineering
  - Weather Stream

Next Generation Backbone Microwave Sounder

- Backbone microwave sounder for the NEON program
- Class C
- Base of 4 instruments, option for an additional 4 instruments
- Planning to include:
  - Bands around all Advanced Technology Microwave Sounder channels plus F band and J band
  - Radio Frequency Interference detection and hyperspectral capabilities
  - Intercalibration performance
- Completed user community white papers, Formulation Studies and Request For Information responses informed requirements

Draft RFP Information

- Four documents updated since the Request For Information (RFI)
  - Statement of Work (SoW)
  - Contract Data Requirements List (CDRL)
  - Performance Requirements Document (PRD)
  - Instrument Mission Assurance Requirements (IMAR)
- New documents for the draft Request For Proposal (RFP)
  - Proposal Instructions
  - Evaluation Criteria
  - Sample Contract
- Responses due on May 9th
- One on ones welcome and can continue past May 9th up until the blackout period beginning at RFP release

Technical Highlights

- Changes in which bands require Radio Frequency Interference detection and hyperspectral capabilities (PRD Table 4.1-2)
- Specific heritage channels are now required (PRD Table 4.1-3)
  - There is flexibility on what the additional channels are
- Noise Equivalent Delta Temperature (NEDT) is now scaled with bandwidth and integration time (Table 4.1-4)
- A Radio Frequency Interference detection section has been added to the PRD (Section 4.2)
- Pointing requirements have been added to the PRD (Table 5.4.2-1)
- Environmental requirements like Electro Magnetic Interference have been refined (PRD Section 6)
- The ICD and MICD will be coordinated with the spacecraft Industry Partner
- The PRD has been renumbered and reorganized
  - Prior requirement IDs may not align with these requirement IDs

Programmatic Highlights

- The Engineering Test Unit is required to be built, assembled, and functionally tested by CDR (SoW Section 4.3.1)
  - Environmental testing is not required before CDR
- SMBA will implement Class C as a mixture of insight and oversight
  - The NEON Team gains insight through partnership relationship with Industry Partners
    - Reliance on empowered Resident Office and Industry Partner relationships
  - Fewer deliverables in the CDRL
    - No "review" deliverables, mostly "information" deliverables, with targeted "approval" deliverables
  - Fewer SoW requirements
  - More decisional authority given to Industry Partner

Role of Industry in Series-1 and SMBA

- All architectural elements acquired commercially!
  - Instrument — Industry Development
  - Spacecraft
  - Launch vehicle
  - Mission operations
  - Data Acquisition and Routing

  (Spacecraft, Launch vehicle, Mission operations, Data Acquisition and Routing: Commercial Mission Services)
- Intercalibration targets are best provided by the Government
  - Enables Industry Partners

Requested Feedback

1. Which changes would you like to see? Which changes would add efficiency? Why?
2. Are there any items you would prefer remain unchanged? If so, what are they and why?
3. Specifically, how could a Class C approach be tailored to allow for greater use of your products and processes? Which requirements could be replaced with your products/processes?
4. How should interfaces be specified that allow for compatibility with as many spacecraft as possible? E.g., what if Ethernet is chosen as a data interface? Are requirements related to the thermal interface specified correctly? Etc. How would you specify these requirements?
5. How would a Fixed Price contract type change your proposal and implementation approach? Please provide pros and cons for cost plus and fixed price contract types.
6. What else would you like us to know?

We sincerely welcome feedback and suggestions!

Redlines on everything are both welcomed and encouraged

Current Notional Implementation Phase Timeline

- RFI Release – 4/17/24
- RFI Response Due – 5/2/24
- Draft RFP Release – 4/18/25
- Draft RFP Responses Due – 5/9/25
- RFP Release – Q4 CY25
- Award – Q4 CY26

Questions?